

ECON 108

INTRODUCTION TO ECONOMICS

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ELASTICITY

CHAPTER IX (from the book)

CHAPTER OUTLINE

Price Elasticity of Demand

- Determinants
- Computation
- Extreme cases
- Relationship with the Slope and with Total Revenues

Other Demand Elasticities

- Income Elasticity
- Cross-Price Elasticity

CHAPTER OUTLINE

Price Elasticity of Supply

- Determinants
- Computation
- Supply Equation and Elasticity

Demand Elasticity, Supply Elasticity, and Volatility

PRICE ELASTICITY OF DEMAND

Elasticity

Measures the responsiveness of buyers and sellers to changes in market conditions.

PRICE ELASTICITY OF DEMAND

Price Elasticity of Demand

The law of demand defined the inverse relationship between the price of a good and the quantity demanded of that good (as price goes up, quantity demanded goes down and vice versa)

PRICE ELASTICITY OF DEMAND

Price Elasticity of Demand

Elasticity allows us to measure how much the quantity demanded changes in response to a change in price.

PRICE ELASTICITY OF DEMAND

Price Elasticity of Demand

The **price elasticity of demand** measures the responsiveness of quantity demanded of a commodity to a change in the price of the commodity.

PRICE ELASTICITY OF DEMAND

Price Elasticity of Demand

If the quantity demanded changes significantly as a result of a price change, then demand is **elastic**. If the quantity demanded changes by a small amount as a result of a price change, then demand is **inelastic**.

PRICE ELASTICITY OF DEMAND

Determinants of the Price Elasticity of Demand

- The existence of substitutes
- The share of the budget spent of the good
- Time to adjust

PRICE ELASTICITY OF DEMAND

Determinants of the Price Elasticity of Demand: The Existence of Substitutes

The larger the number of available substitutes the more elastic demand is to changes in prices.

PRICE ELASTICITY OF DEMAND

Determinants of the Price Elasticity of

Demand: The Share of the Budget Spent on the Good

The larger the share of the budget spent on the good, the more elastic demand is to changes in prices.

PRICE ELASTICITY OF DEMAND

Determinants of the Price Elasticity of

Demand: Time to Adjust

- **The immediate-run:** a period during which there is no time to adjust behavior.
- **The short-run:** a period during which there is some flexibility allowing some change in behavior.
- **The long run:** a period during which there is complete flexibility to adjust behavior.

PRICE ELASTICITY OF DEMAND

Determinants of the Price Elasticity of Demand: Time to Adjust

Demand is more elastic in the long run than in the short run, and it is perfectly inelastic in the immediate run.

PRICE ELASTICITY OF DEMAND

Computing the Price Elasticity of Demand

$$E_p = \frac{\text{Percentage change in quantity demanded}}{\text{Percentage change in price}}$$

PRICE ELASTICITY OF DEMAND

Computing the Price Elasticity of Demand

Example:

- Price of oil increases 10%
- Quantity demanded decreases 1%

$$E_p = \frac{-1\%}{+10\%} = -.1$$

PRICE ELASTICITY OF DEMAND

Computing the Price Elasticity of Demand

Question:

How would you interpret an elasticity of -0.1 ?

Answer:

A 10% increase in the price of oil will lead to a 1% decrease in quantity demanded.

PRICE ELASTICITY OF DEMAND

Computing the Price Elasticity of Demand

Observations:

- Elasticity is concerned with percentage changes.
- An increase in price decreases the quantity demanded (*ceteris paribus*) and hence price elasticity of demand is always negative.
- By convention, the minus sign is ignored (we look at the **absolute value** of the price elasticity of demand).

PRICE ELASTICITY OF DEMAND

Computing the Price Elasticity of Demand

Elastic Demand: $|E_p| > 1$

Percentage change in quantity demanded is larger than the percentage change in price.

PRICE ELASTICITY OF DEMAND

Computing the Price Elasticity of Demand

Inelastic Demand: $|E_p| < 1$

Percentage change in quantity demanded is smaller than the percentage change in price.

PRICE ELASTICITY OF DEMAND

Computing the Price Elasticity of Demand

Unitary or Unit-Elastic Demand: $|E_p| = 1$

Percentage change in quantity demanded is equal to the percentage change in price.

PRICE ELASTICITY OF DEMAND

Computing the Price Elasticity of Demand

There are two ways of measuring the price elasticity of demand:

- At a particular point (point elasticity)
- Between two points (Arch elasticity or midpoint elasticity)

PRICE ELASTICITY OF DEMAND

Extreme Cases

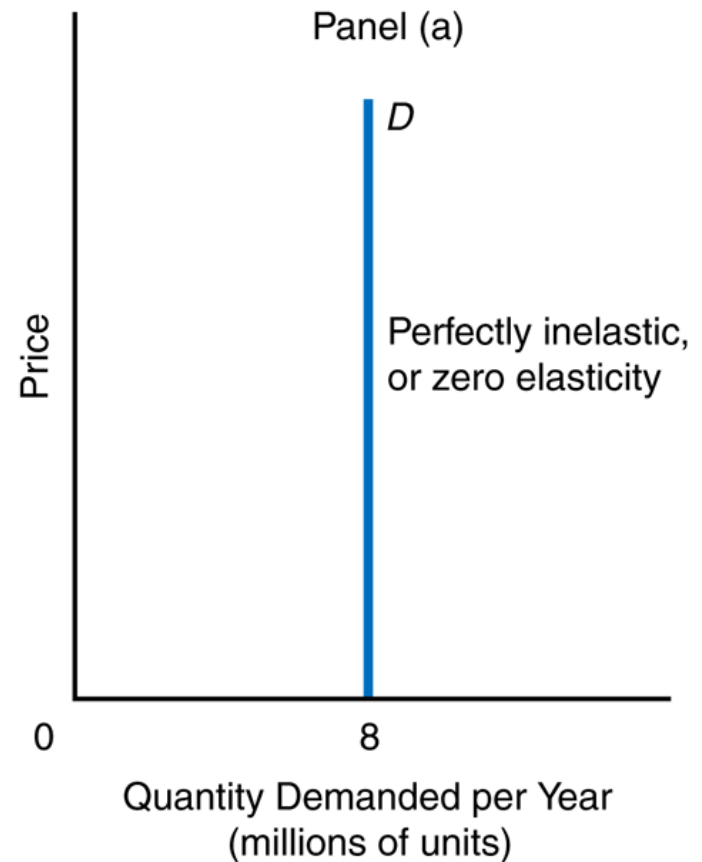
Perfectly Inelastic Demand

- A demand curve that is a vertical line
- It has only one quantity demanded for each price.
- No matter what the price, quantity demanded does not change.
- A demand that exhibits zero responsiveness to price changes.

PRICE ELASTICITY OF DEMAND

Extreme Cases

Perfectly Inelastic Demand



PRICE ELASTICITY OF DEMAND

Extreme Cases

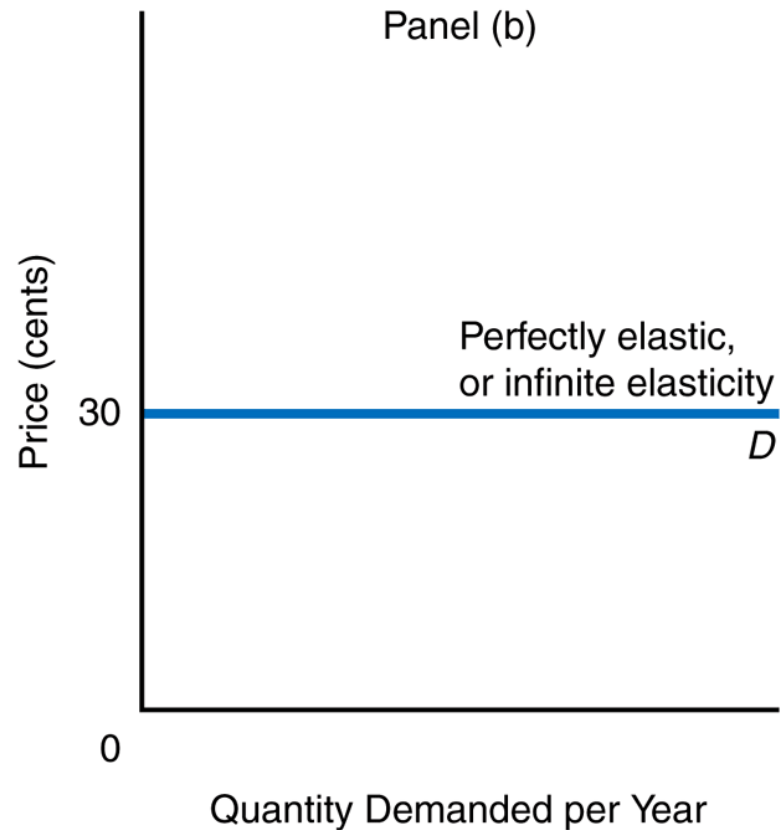
Perfectly Elastic Demand

- A demand curve that is a horizontal line
- It has only one price for every quantity.
- The slightest increase in price leads to zero quantity demanded.

PRICE ELASTICITY OF DEMAND

Extreme Cases

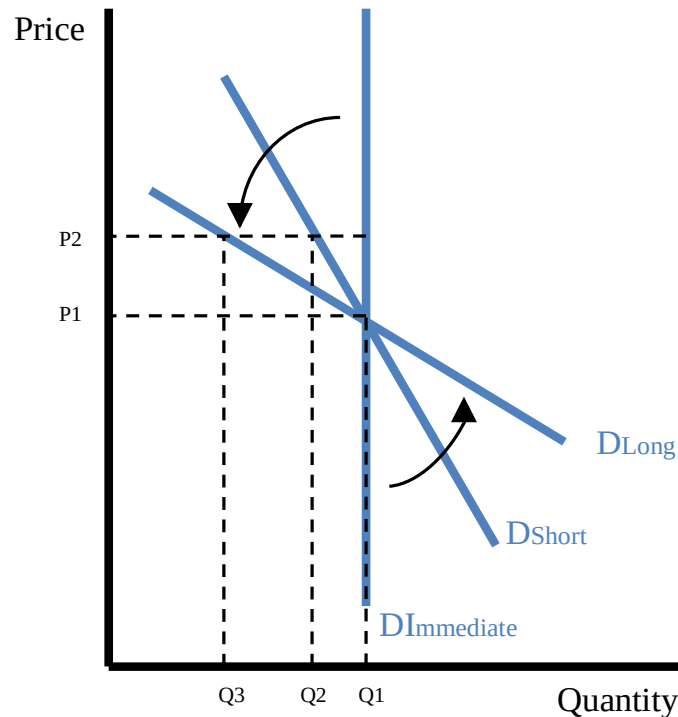
Perfectly Elastic Demand



PRICE ELASTICITY OF DEMAND

Time, Elasticity, and the Demand Curve

- Increased time makes demand more elastic.



PRICE ELASTICITY OF DEMAND

Slope and Elasticity

- Recall that the slope is the rate of change in the y-axis divided by the rate of change on the x-axis—the rise over the run.
- The elasticity formula is the *percentage change* along the x-axis divided by the *percentage change* along the y-axis.
- Even if the slope is constant in a linear demand curve, elasticity is not!

PRICE ELASTICITY OF DEMAND

Price Elasticity of Demand and Total Revenues

- When price elasticity of demand is more than one (elastic) lowering the price will increase total revenue.
- When price elasticity of demand is one (unit elastic) lowering the price will not change total revenue.
- When price elasticity of demand is less than one (inelastic) lowering the price will decrease revenue.

PRICE ELASTICITY OF DEMAND

Price Elasticity of Demand and Total Revenues

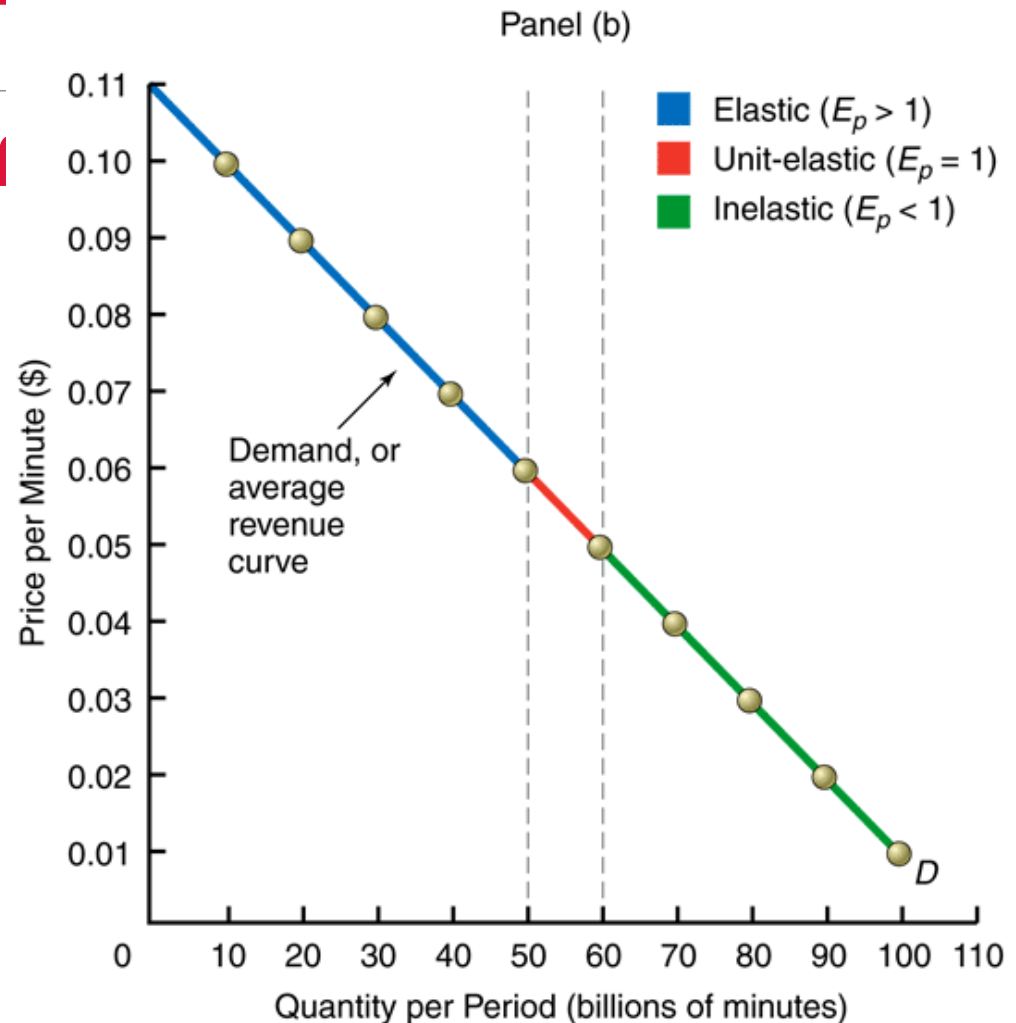
Panel (a)

(1) Price, P , per Minute of Cellular Phone Service	(2) Quantity Demanded, Q (billions of minutes)	(3) Total Revenue (\$ billions) = (1) x (2)	(4) Elasticity, $E_P = \frac{\text{Change in } Q}{(Q_1 + Q_2)/2} \div \frac{\text{Change in } P}{(P_1 + P_2)/2}$
\$0.11	0	0	
0.10	10	1.0	21.000
0.09	20	1.8	6.330
0.08	30	2.4	3.400
0.07	40	2.8	2.143
0.06	50	3.0	1.144
0.05	60	3.0	1.000
0.04	70	2.8	.692
0.03	80	2.4	.467
0.02	90	1.8	.294
0.01	100	1.0	.158

{ Elastic
{ Unit-elastic
{ Inelastic

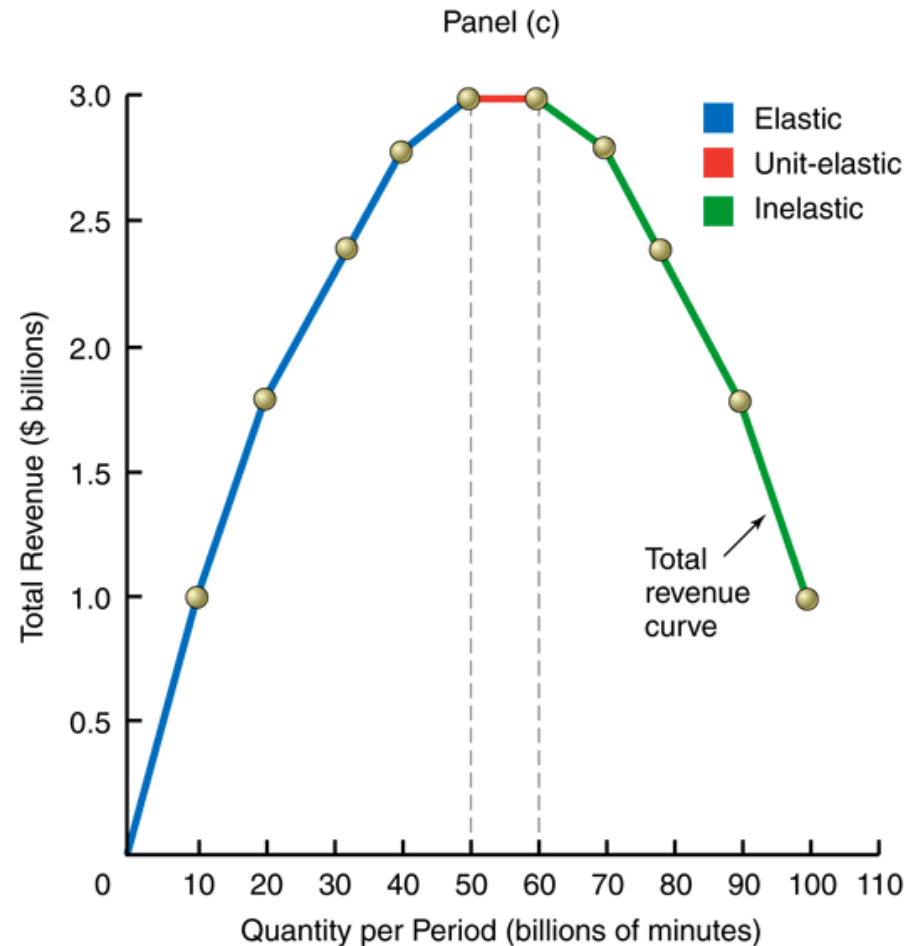
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Price Elasticity of Demand and Total Revenues



PRICE ELASTICITY OF DEMAND

Price Elasticity of Demand and Total Revenues



PRICE ELASTICITY OF DEMAND

Price Elasticity of Demand and Total Revenues

Price Elasticity of Demand (E_p)		Effect of Price Change on Total Revenues (TR)	
		Price Decrease	Price Increase
Inelastic	$(E_p < 1)$	TR ↓	TR ↑
Unit-elastic	$(E_p = 1)$	No change in TR	No change in TR
Elastic	$(E_p > 1)$	TR ↑	TR ↓

OTHER DEMAND ELASTICITIES

Income Elasticity of Demand

- The responsiveness of demand to changes in income, other things constant.
- The percentage change in demand for any good, holding its price constant, divided by the percentage change in income.

OTHER DEMAND ELASTICITIES

Income Elasticity of Demand

$$E^d_I = \frac{\text{Percentage change in demand}}{\text{Percentage change in income}}$$

OTHER DEMAND ELASTICITIES

Income Elasticity of Demand

$E^d_I < 0$: an increase in income will decrease the quantity demanded of the commodity at each price. The commodity is **inferior**.

$E^d_I \geq 0$: an increase in income will increase the quantity demanded of the commodity at each price. The commodity is **normal**.

- $E^d_I > 1$: The good is superior (luxury)
- $0 \leq E^d_I \leq 1$: The good is a necessity (necessary)

OTHER DEMAND ELASTICITIES

Cross-Price Elasticity of Demand

- The responsiveness of demand to changes in the price of a different good, other things constant.
- The percentage change in the demand for one good (holding its price constant) divided by the percentage change in the price of a related good.

OTHER DEMAND ELASTICITIES

Cross-Price Elasticity of Demand

$$E_{xy} = \frac{\% \text{ Change in demand for good } X}{\% \text{ Change in price of good } Y}$$

OTHER DEMAND ELASTICITIES

Cross-Price Elasticity of Demand

$E_{xy} > 0$: an increase in the price of X will increase the quantity demanded of Y at each price; goods are **substitutes**.

$E_{xy} < 0$ an increase in the price of X will decrease the quantity demanded of Y at each price; goods are **complements**.

$E_{xy} = 0$ a change in the price of X will have no effect on the quantity demanded of Y ; goods are **not related**.

PRICE ELASTICITY OF SUPPLY

Price Elasticity of Supply

- The responsiveness of the quantity supplied of a commodity to a change in its price, other things constant.
- The percentage change in quantity supplied divided by the percentage change in price.

PRICE ELASTICITY OF SUPPLY

Price Elasticity of Supply

$$E_p^s = \frac{\text{Percentage change in quantity supplied}}{\text{Percentage change in price}}$$

PRICE ELASTICITY OF SUPPLY

Perfectly Elastic Supply

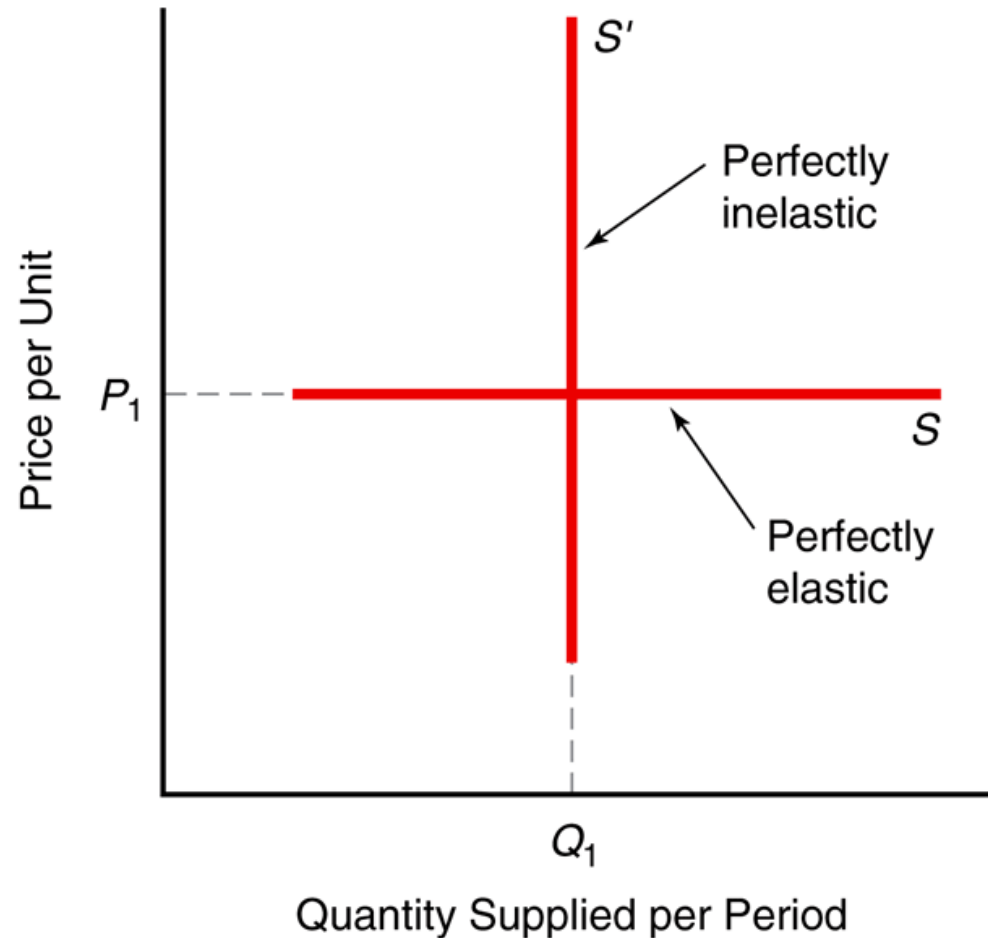
- Quantity supplied falls to zero when there is the slightest decrease in price
- The supply curve is horizontal at a given price

Perfectly Inelastic Supply

- Quantity supplied is constant no matter what happens to price
- The supply curve is vertical at a given price

PRICE ELASTICITY OF SUPPLY

Extreme Cases



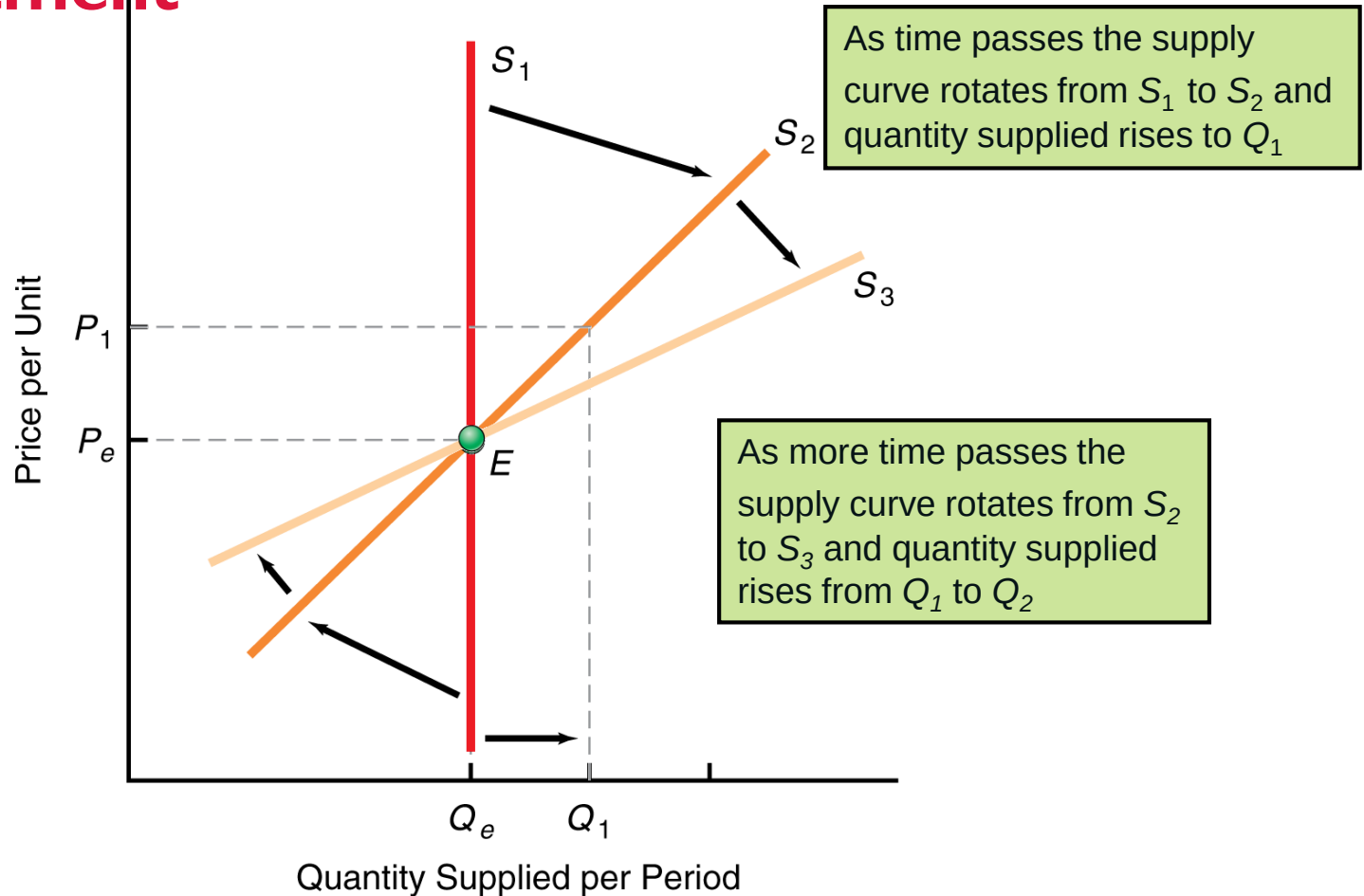
PRICE ELASTICITY OF SUPPLY

Price elasticity of supply and length of time for adjustment

- The longer the time allowed for adjustment, the more resources can flow into (out of) an industry through expansion (contraction) of existing firms.
- The longer the time allowed for adjustment, the entry (exit) of firms increases (decreases) production in an industry.

PRICE ELASTICITY OF SUPPLY

Price elasticity of supply and length of time for adjustment



END OF CHAPTER